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The Ablation Threshold of Er:YAG and Er:YSGG Laser Radiation in Dental Enamel

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Abstract. The scientific investigation of fundamental problems plays a decisive role in understanding the mode of action and the consequences of the use of lasers on biological material. One of these fundamental aspects is the investigation of the ablation threshold of various laser wavelengths in dental enamel. Knowledge of the relationships and influencing factors in the laser ablation of hard tooth tissue constitutes the basis for use in patients and the introduction of new indications. The present paper examines the ablation threshold of an Er:YAG laser ($\lambda = 2.94\ \mu m$) and an Er:YSGG laser ($\lambda = 2.79\ \mu m$) in human dental enamel. To this end, 130 enamel samples were taken from wisdom teeth and treated with increasing energy densities of 2–40 J/cm^2. The sample material was mounted and irradiated on an automated linear micropositioner. Treatment was performed with a pulse duration of $\tau_{P(FWHM)} \approx 150\ \mu s$ and a pulse repetition rate of 5 Hz for both wavelengths. The repetition rate of the laser and the feed rate of the micropositioner resulted in overlapping of the single pulses.

The surface changes were assessed by means of reflected light and scanning electron microscopy. On the basis of the results, it was possible to identify an energy density range as the ablation threshold for both the Er:YAG and the Er:YSGG laser. With the Er:YAG laser, the transition was found in an energy density range of 9–11 J/cm^2. The range for the Er:YSGG laser was slightly higher at 10–14 J/cm^2.

Keywords: Ablation; Ablation threshold; Dental enamel; Er:YAG laser; Er:YSGG laser

INTRODUCTION

The use of lasers in dentistry has acquired increasing importance in recent years. The US Food and Drug Administration (FDA) has approved many different indications and wavelengths for oral laser application [1]. These include approvals for various erbium lasers for the ablation of hard tooth tissue. Knowledge of the relationships and influencing factors in the laser ablation of hard tooth tissue constitutes the basis for use in patients and the introduction of new indications. One of these potential new indications is the use of lasers in caries prevention. Various studies have already been able to demonstrate the potential of laser radiation of different wavelengths for increasing the resistance of dental enamel to exposure to acid [2,3]. Among others, the types of laser tested include the Er:YAG laser (2.94 μm) and the Er:YSGG laser (2.79 μm) [4,5]. As the purpose of this application is not to ablate the dental enamel, but rather to induce a chemical or structural change, accurate knowledge of the ablation threshold of these laser wavelengths constitutes the basis for their use.

Enamel is a biological composite containing by volume, 8–12% water, 85–95% mineral, and 2–3% protein and lipid. These components are not evenly distributed throughout the tissue [6]. Erbium lasers can be used to ablate enamel due to the strong water and OH^- absorption of apatite. Er:YSGG laser emission is coincident with the narrow (OH^-) apatite absorption band at $\lambda = 2.8\ \mu m$, whereas Er:YAG emission overlaps the board water absorption centred at $\lambda = 3.0\ \mu m$. The room temperature absorption coefficient of enamel at 2.79 μm ($\mu_a = 400\ cm^{-1}$) is lower by a factor of two vs. 2.94 μm ($\mu_a = 800\ cm^{-1}$), but is expected to change markedly during laser irradiation due

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to the changes in the optical properties of water with increased temperature. Therefore, the average absorption coefficient during laser irradiation may actually be higher at 2.79 μm [7]. Based on the absorption properties of enamel it is accepted that the highly efficient thermomechanical ablation process of the erbium lasers is initiated by heating of tissue water. During rapid heating, the tissue water can create enormous subsurface pressure that can lead to the explosive removal of the surrounding mineral matrix [7–9].

The present study investigates the ablation threshold of Er:YAG and Er:YSGG laser radiation in the dental enamel of human teeth.

MATERIAL AND METHOD

The sample material consisted of human retained wisdom teeth with completed root growth that needed to be removed by extraction or osteotomy for medical reasons. After extraction, the teeth were cleaned and stored in 0.9% physiological saline solution (NaCl) for no more than 1 week. The teeth were halved in the axial direction in order to carry out the experiments. A total of 130 samples were produced. The entire area of the enamel was irradiated, starting in the cervical region and proceeding via the buccal or oral central surface to the occlusal region. This made it possible to cover areas of the dental crown enamel with different degrees of mineralisation. In order to avoid any secondary effects caused by additional use of a cooling mechanism, air or water cooling was deliberately dispensed with during irradiation.

The tests were performed using a Type 1-2-3 modular, experimental laser system from Schwartz Electro-Optics. This system offers the option of exchanging the laser crystals, making it possible to emit light with a wavelength of $\lambda = 2.94$ μm for the Er:YAG laser and $\lambda = 2.79$ μm for the Er:YSGG laser. Both erbium lasers were operated in free-running mode. They display the typical pulse pattern of free-running solid-state lasers (spiking), i.e. a series of individual pulses with a pulse width τ_P of roughly 1 μs. The total pulse duration of both lasers is in the region of 200–250 μs (where $\tau_{P(FWHM)} \approx 150$ μs; FWHM = (full width at half maximum)).

The tooth samples were irradiated with increasing energy densities. Energy densities of 2–20 J/cm^2 were selected for the Er:YAG laser and 4–40 J/cm^2 for the Er:YSGG laser. The laser was used with a pulse repetition rate of 5 Hz for both wavelengths.

Laser irradiation was performed at constant, maximum output, in order to prevent the beam properties from being affected by changes in the thermal lens within the laser crystal. The pulse energy was regulated with the help of the absorber plates positioned behind the laser. In order to avoid effects of the transversal mode profile, which manifests itself in the form of inhomogeneous distribution of the laser energy on the sample surface, the laser light was coupled into a fluoride glass fibre with a length of 1.2 m. This made it possible to guarantee virtually gaussian distribution of the emerging light. The light hit the tooth sample downstream of the fibre ($Ø_{core} = 535$ μm, NA = 0.14), the light guide being protected against contamination with ablated tooth particles by a further, transparent plate. The samples were moved perpendicularly through the laser beam on a computer-controlled linear micropositioner, so as to obtain a linear incision on the tooth surface (Fig. 1).

The teeth were irradiated immediately after removal from the storage solution, in order to avoid drying of the hard tooth tissue and resultant falsification of the results.

The beam path was fitted with a beam splitter, so that the energy could be measured (two-channel energy meter and detectors by Laserprobe) simultaneously during each irradiation process. The energy density of the laser beam at the location of the sample was determined in every second measurement. This was achieved using the 'knife-edge' method, where a razor blade is moved perpendicularly through the beam on the micropositioner. The energy profile behind the blade is recorded by a detector. The first differentiation of the measured curve corresponds to the profile of the laser beam (determined at $1/e^2$ intensity level), provided that gaussian distribution and radial symmetry are ensured. Both prerequisites are met by using a fluoride glass fibre. Figure 2 shows the measurement of the beam profile in the plane of irradiation. This made it possible to determine the energy density of the laser beam on the surface of the sample for use in determining the ablation threshold values.

The selected pulse repetition rate of the laser of 5 Hz, in conjunction with the feed rate of the linear micropositioner of 0.022 cm/s

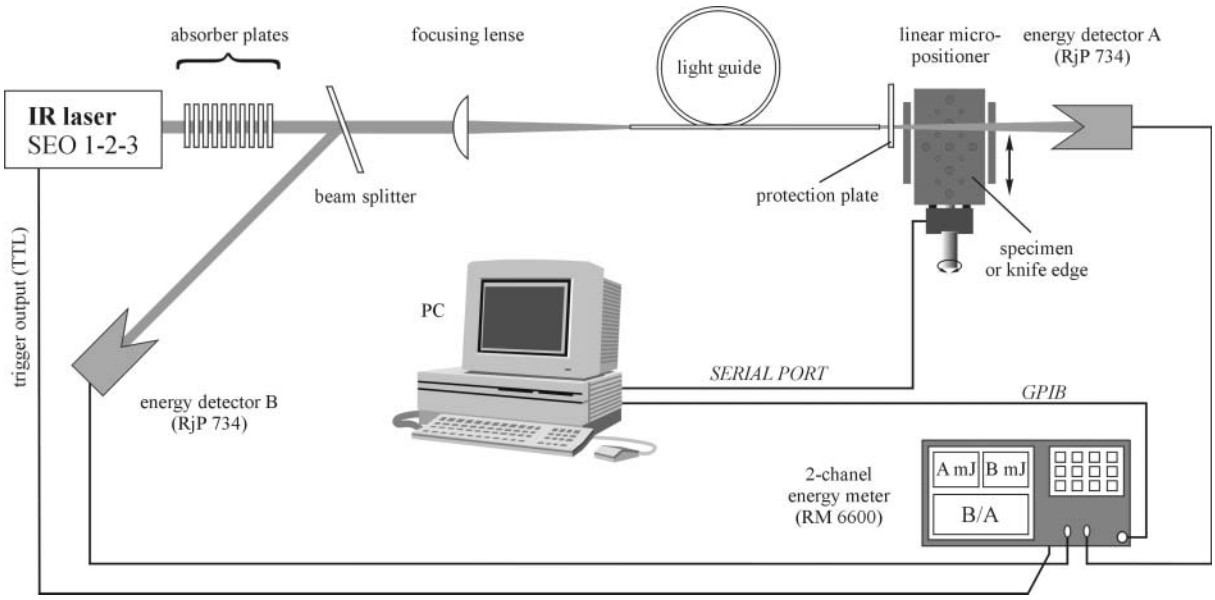


Fig. 1. Experimental set-up.

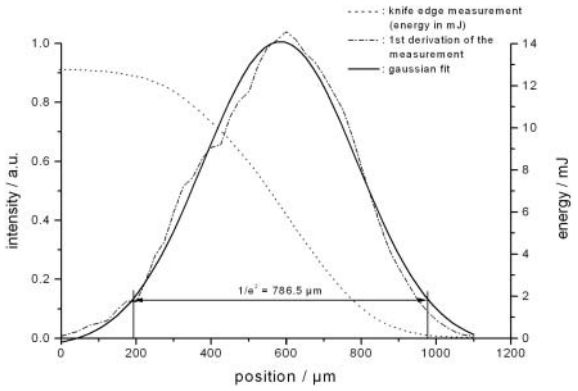


Fig. 2. Knife edge measurement, first derivation and gaussian fit of the beam profile in the plane of irradiation. A beam diameter of 786.5 µm was calculated at the intensity level of $1/e^2$.

($\Delta x = 44$ µm) and the beam diameter of ≈ 800 µm on the tooth surface (Fig. 2), resulted in an overlap of the individual pulses n of $n=9$ [10,11], meaning that almost 9 pulses are placed in the space of a pulse radius.

A note was made of visual observations made during and after laser irradiation, such as discoloration of the hard tooth tissue, sparking or smoke evolution. After irradiation, we used a Type 475057 stereomicroscope from Zeiss to examine the samples for signs of ablation under a reflected-light microscope. Detailed observation of the irradiated areas was then performed with the help of scanning electron microscope images (Type: LEO DSM 962).

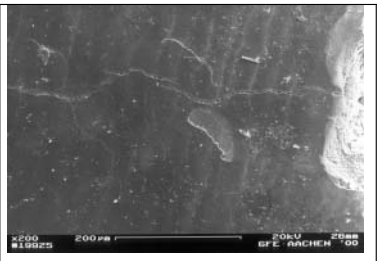
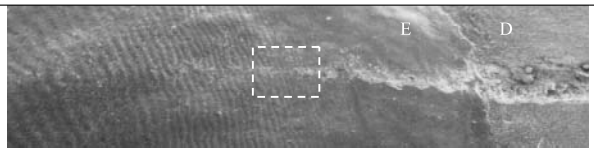
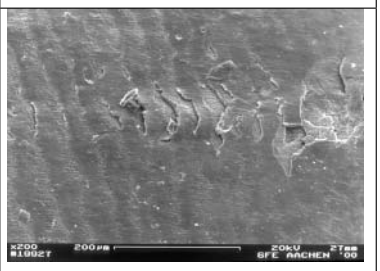
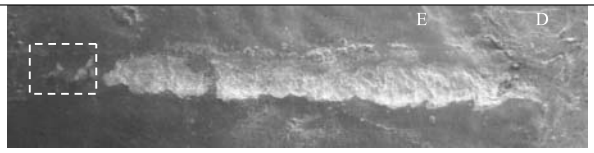
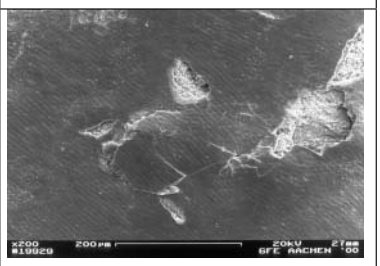
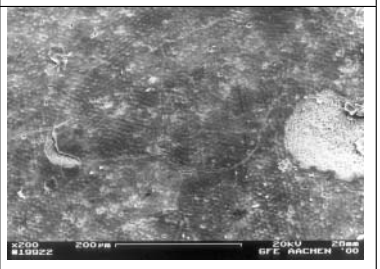
RESULTS

Based on observations under a reflected-light microscope and images obtained with a scanning electron microscope (Figs 3 and 4), we were able to define energy density ranges as ablation thresholds for the Er:YAG and Er:YSGG lasers. With the Er:YAG laser, the transition was found to lie in the range of energy densities between 9 and 11 J/cm^2. In a few exceptional cases, however, ablation already occurred below this threshold or, conversely, no ablation was seen above this threshold.

The range for the Er:YSGG laser was slightly higher at 10–14 J/cm^2. Again, there were some 'mavericks' below and above this range.

In addition, there was found to be a marked difference in the ablation behaviour of both wavelengths in the various regions of the dental crown. Ablation occurred at an early stage in the region of the cervical enamel, decreasing across the central buccal or oral surface towards the occlusal surface. This effect was particularly noticeable with the Er:YSGG laser.

Moreover, the scanning electron microscope images of the Er:YSGG wavelength show microcracks in the outermost, prism-free area of the enamel after irradiation. In some cases, these cracks even occur before any noticeable ablation (Fig. 3).

Energy density	Plain image (12x)	Magnification (200x)
8 J/cm²	Ablation in the dentin and in the transition to the enamel area. No subsequent ablation in the dental enamel. Light appearance of the irradiated area with occurrence of microcracks.	
9 J/cm²	 Ablation in the dentin and in the neck area of the tooth. Less or no ablation on the central, buccal tooth surface, isolated cracking of the upper enamel layer	
12.6 J/cm²	 Ablation over the entire irradiated area.	
19.5 J/cm²	Ablation in the neck area of the tooth and on the buccal surface. Light appearance of the irradiated area with occurrence of microcracks.	

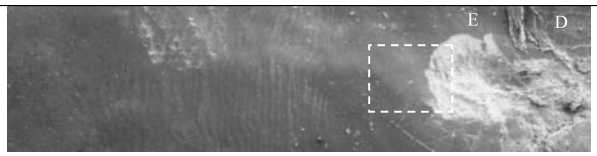
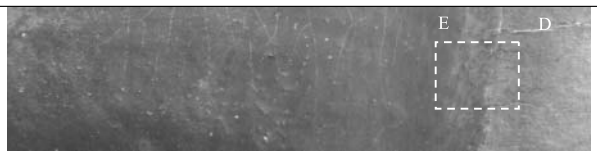
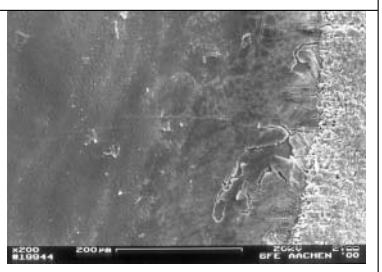
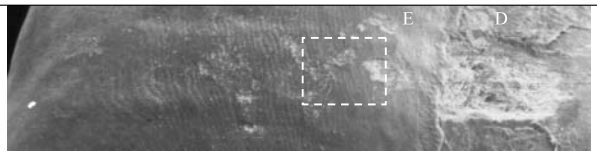
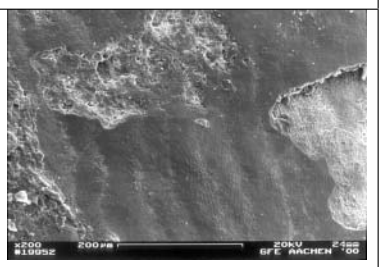
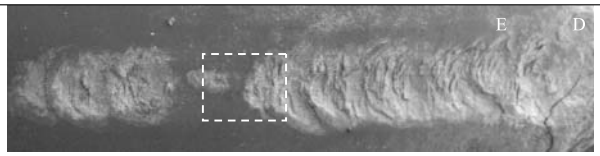
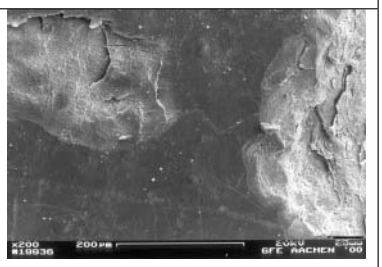
* E = enamel, D = dentin / cementum

Fig. 3. Sample scanning electron microscope images for the Er:YSGG laser.

Slight smoke evolution and faint, white or brownish discoloration within the dental enamel were observed in some cases during irradiation of the enamel samples with both laser wavelengths. The occurrence of these effects was irregular and could not be assigned to any particular energy density.

Statistical Analysis

The results were analysed statistically by calculating the probability of the occurrence of ablation for the Er:YAG and Er:YSGG laser with the help of logistic regression. Figure 5 shows the probability of the occurrence of ablation as a function of the energy density.

Energy density	Plain image (12x)	Magnification (200x)
5.3 J/cm^2	 Ablation in the dentin and in the transition to the dental enamel. No ablation in the further course of irradiation.	
8.6 J/cm^2	 No ablation in the dental enamel. Cracking in the adjacent dentin as a result of dehydration.	
11.2 J/cm^2	 Ablation in the dentin and in the area of the cervical enamel.	
13.7 J/cm^2	 Ablation over the entire irradiated area.	

* E = enamel, D = dentin / cementum

Fig. 4. Sample scanning electron microsope images for the Er:YAG laser.

The sensitivity and specificity of the test were also determined. Based on a sensitivity of 80%, the ablation thresholds for the Er:YAG and Er:YSGG lasers can be defined as 10 J/cm^2 and 13 J/cm^2. This means it can be stated with 80% certainty that, in our test, ablation occurred at 10 J/cm^2 with the Er:YAG laser and 13 J/cm^2 with the Er:YSGG laser.

The specificity is in the region of 80% for both erbium lasers, putting it in the same order of magnitude as the sensitivity. Consequently, 20% of all irradiated samples fail to provide any proper results at these energy densities. In these cases, either ablation already occurred below these threshold values, or there was no ablation above the threshold.

The gradient of the fit functions shows the fluctuation of ablation threshold. The steep slope for the erbium:YAG laser implements a sharper cut off point where ablation occurs.

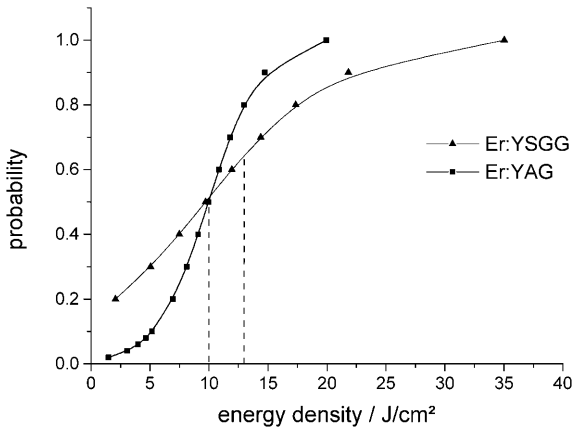


Fig. 5. Probability of the occurrence of ablation as a function of the energy density.

The transition to ablation broadens with decreasing gradient of the propability fit functions. This corresponds to the observated ablation range of the erbium:YSGG laser.

DISCUSSION

The introduction of erbium lasers in dentistry initially pursued the objective of ablating hard tooth tissue. Knowledge of the relationships and influencing factors in the laser ablation of hard tooth tissue constitutes the basis for use in patients and the introduction of new indications. One of these potential new indications is the irradiation of dental enamel below the ablation threshold in order to increase its resistance to acid. When using laser systems that emit wavelengths in the region of 3 μm, this necessitates precise knowledge of the ablation threshold and the factors influencing it. The dental enamel is not to be ablated, but merely undergo chemical or structural change.

The present paper investigates the ablation threshold of Er:YAG and Er:YSGG lasers in dental enamel. Only human enamel samples were used. The outermost, prism-free enamel layer was cleaned, but not polished. The enamel samples were stored in physiological saline solution to prevent dehydration, as the water in the dental enamel is the actual absorber of the erbium laser radiation and thus the mediator of the thermomechanical ablation process [7–9].

The ablation thresholds determined, i.e. 10 J/cm^2 for the Er:YAG laser and 13 J/cm^2 for the Er:YSGG laser, are not exact thresholds and can only be regarded as ranges of transition to ablation. In this context, we defined a sensitivity of 80% in the statistical analysis. This means that a false-negative result is permitted in 20% of the irradiated samples. The reason for this lies in the fluctuations that arise when using biological material. Even slight differences in the degree of mineralisation of the dental enamel have an effect on ablation.

The results obtained for the Er:YAG laser correlate very well with the information in the literature. In their paper, Fried et al. [4] define the ablation threshold as being 7–9 J/cm^2. Hibst and Keller [9] state approx. 10 J/cm^2 as the ablation threshold for the Er:YAG laser. Belikov et al. [12] found the ablation threshold at 8 J/cm^2 in their study.

For the Er:YSGG laser, the present paper determined lower threshold values than given in the paper by Fried et al. [4]. This could possibly be explained by the fact that different areas of the dental crown were irradiated. Although a comparable set-up was used in this study, only the central, buccal area of the coronal enamel was irradiated. In their study, Belikov et al. [12] determined an ablation threshold of approx. 4 J/cm^2 for the Er:YSGG laser. However, they used an experimental set-up with a pulse length of 400 ms for their work. Consequently, the results cannot be applied to the present study.

In determining the ablation threshold, none of the previous papers consider differences in the composition of the hard tooth tissue in the various areas of the dental crown.

The results of the present study clearly illustrate that the exact determination of the ablation thresholds is very difficult and influenced by a variety of different factors. For example, the physical influencing factors include the pulse length, as well as the transversal mode profile of the laser. The results obtained here refer exclusively to Er:YAG and Er:YSGG lasers with a pulse length of $\tau_{P(FWHM)} \approx 150$ μs. It must be borne in mind that the results are only valid for virtually gaussian beam distributions.

Owing to their resonator geometry, most erbium lasers do not generate a TEM_{00} mode profile (transversal excited mode, 00 = gaussian beam profile). This multimode profile structure also remains largely unaffected by the use of articulated mirror arms and tips, meaning that it is impossible to make an accurate statement concerning the emitted energy density. When using an optical light guide of suitable length, in particular, the mode structures are

destroyed by multiple reflection, thus achieving a nearly gaussian distribution of the light at the distal end of the light guide.

Comparing the width of ablation grooves in Figs 3 and 4 with the diameter of the used fibre it is ocularly that the size of the grooves is much smaller than the spot of irradiation ($\approx 800\ \mu m$). Due to the diameter of the light guide combined with its gaussian beam profile, the energy distribution is non-uniform over the whole area (Fig. 2). The energy density in the centre is higher than in the margin. Therfore ablation occurs in the beam centre first.

Additional influencing factors arise from the use of dental enamel as the study material. As usual with biological material, there are also regional differences in the composition of dental enamel [6]. Even slight variations in water content lead to different results. In the present study, for example, differences in ablation behaviour were observed between the cervical and buccal or oral regions of the enamel. Ablation occurs at an early stage in the cervical region as a result of its lesser degree of mineralisation. It is not yet clear why this effect is more noticeable with the Er:YSGG laser than with the Er:YAG laser. One possibility is that the slight differences in the absorption behaviour of the two wavelengths are equalised as the water content of the hard tooth tissue increases.

In addition to ablation, various other effects were also observed. For instance, slight smoke evolution and opaque, chalky or brownish discoloration within the dental enamel occurred even below the ablation threshold in some cases. However, these effects occurred in irregular fashion and could not be assigned to any specific energy density or wavelength. At the wavelength of the Er:YSGG laser, there is a possibility of microcracks developing in the dental enamel below the ablation threshold. Whether and to what extent these effects have an impact on the structure and mechanical properties of the dental enamel remains to be clarified in further studies.

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REFERENCES

1. Lasers in Dentistry. www.fda.gov.
2. Featherstone JDB, Barrett-Vespone NA, Fried D, Kantorowitz Z, Seka W. CO_2 laser inhibition of artificial caries-like lesion progression in dental enamel. J Dent Res 1998;77:1397–403.
3. Powell GL, Higuchi WI, Fox JL, Yu D, Blankenau R. Argon laser effect on demineralisation of human enamel. Proc SPIE 1992;1643:374–9.
4. Fried D, Featherstone JDB, Visuri SR, Seka W, Walsh JT. The caries inhibition potential of Er:YAG and Er:YSGG laser radiation. Proc SPIE 1997;2672:73–8.
5. Apel C, Gräber HG, Gutknecht N. Calcium solubility of dental enamel following Er,Cr:YSGG laser irradiation. Proc SPIE 2000;3910:318–21.
6. Schröder HE. Orale Strukturbiologie, 4th edn, Stuttgart, New York: Thieme, 1992.
7. Fried D. IR laser ablation of dental enamel. Proc SPIE 2000;3910:136–48.
8. Majaron B, Lukač M. Thermo-mechanical laser ablation of hard dental tissues: an overview of effects, regimes and models. Proc SPIE 1999;3593:184–95.
9. Hibst R, Keller U. Experimental studies of the application of the Er:YAG laser on dental hard substances: I. Measurement of the ablation rate. Lasers Surg Med 1989;9:338–44.
10. Ivanenko MM, Hering P. Hard tissue ablation with a mechanically Q-switched CO_2 laser. Proc SPIE 1998; 3565:110–15.
11. Stanislawski M, Meister J, Mitra T, Ivanenko MM, Zanger K, Hering P. Hard tissue ablation with a free-running Er:YAG and a Q-switched CO_2 laser: a comparative study. Appl Phys B 2001;72:115–20.
12. Belikov AV, Erofeev AV, Shumilin VV, Tkachuk AM. Comparative study of the 3 μm laser action on different tooth tissue samples using free-running Er-doped YAG, YSGG, YAP and YLF lasers. Proc SPIE 1993; 2080:60–7.

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